

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate Safety and Efficacy of Ocrelizumab in Comparison With Fingolimod in Children and Adolescents With Relapsing-Remitting Multiple Sclerosis (RRMS)
Short Title/Acronym: OPERETTA 2 **Protocol Number:** WN42086 **NCT Number:** NCT05123703 **Posting ID:** 920486564724314556 **Sponsor Name:** Roche **Company:** Roche **Investigational Product:** Ocrelizumab (Ocrevus) vs. Fingolimod (Gilenya, Tascenso) **Phase of Development:** Phase III **Trial Status:** Active, Not Recruiting
Medical Monitor: Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of ocrelizumab compared with fingolimod in terms of controlling clinical relapses in children and adolescents with Relapsing-Remitting Multiple Sclerosis (RRMS). **Secondary Objectives:** To evaluate the effect of ocrelizumab on MRI endpoints (such as new or enlarging T2 lesions and T1 gadolinium-enhancing lesions) and to assess its long-term safety and tolerability profile in the pediatric population.

2.2 Background & Rationale Pediatric-onset MS (POMS) is an often highly active disease that accounts for 3% to 5% of all MS cases worldwide. Preventing disease activity, relapses, and the accrual of irreversible neurological impairment is a key goal. Currently, almost half of children and young people stop MS treatment due to side effects or a lack of efficacy, underscoring an urgent need for more disease-modifying therapy options. This study is conducted to determine if ocrelizumab (an established adult MS therapy) can safely achieve noninferiority or superiority against fingolimod in children to attain complete remission of clinical and MRI activity.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Fingolimod) **Blinding:** Double-blind, Double-dummy **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 188 participants (originally planned at 171) **Number of Sites:** Approximately 103 locations globally **Estimated Study Start:** 19-May-2022 **Estimated Primary Completion:** 09-Jun-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age between 10 and < 18 years. 2. Body weight $\geq$ 25 kilograms (kg). 3. Confirmed diagnosis of RRMS in accordance with the International Pediatric Multiple Sclerosis Study Group (IPMSSG) criteria for pediatric Multiple Sclerosis (MS), Version 2012, or McDonald criteria 2017. 4. Expanded Disability Status Scale (EDSS) at screening: 0-5.5, inclusive. 5. For all countries except Germany, at least one MS relapse during the previous year or two MS relapses in the previous 2 years, or evidence of at least one Gd enhancing lesion on MRI within 6 months. 6. **Inclusion Criteria for Optional OLE Period:** Participants in Group A (ocrelizumab in the double-blind period [DBP]) and Group B (fingolimod in the DBP) who, in the opinion of the investigator, may benefit from switching to ocrelizumab and who have completed the DBP with study treatment (ocrelizumab/fingolimod), may participate in the OLE period.

4.2 Exclusion Criteria 1. Known presence or suspicion of other neurologic disorders that may mimic MS. 2. Significant uncontrolled somatic diseases, known active infection, or any other significant condition that may preclude the participant from participating in the study. 3. Participants with severe cardiac disease or significant findings on the screening Electrocardiograph (ECG). 4. **Exclusion Criteria for Optional OLE Period:** Participants who have discontinued the study during the DBP.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Ocrelizumab:** Administered via intravenous (IV) infusion every 24 weeks (Q24W). The first dose is given as dual infusions of half the dose on Days 1 and 15. Dosing is weight-adjusted: 300 mg for participants weighing < 35 kg, and 600 mg for participants weighing $\geq$ 35 kg. * **Fingolimod:** Administered orally as a once-daily (QD) capsule. * **Placebo:** Administered in a double-dummy design (either IV ocrelizumab placebo or oral fingolimod placebo). **Route of Administration:** Intravenous (IV) and Oral (PO). **Duration of Treatment:** Flexible duration for the double-blind period (lasting until the last randomized participant has completed 24 weeks), followed by an optional open-label extension (OLE) period of at least 144 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed therapies for managing MS symptoms. **Prohibited:** Concurrent use of other disease-modifying therapies (DMTs) for MS outside of the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, ECG, MRI, Baseline Labs
Baseline	Day 1	Randomization, First Dose (First half IV infusion + PO capsule)
Interim 1	Day 15	Second half-dose of the first IV infusion cycle
Follow-Up	Weeks 12, 24	Efficacy Assessment (MRI, Relapse Evaluation), Safety Labs, PO medication restock
End of Study	Double-Blind End	Final Double-Blind Evaluations, Transition to Open-Label Extension (OLE)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized Relapse Rate (ARR) over the treatment period. **Measurement Method:** Clinical assessment of protocol-defined relapses.

7.2 Secondary & Exploratory Endpoints * Number of new or enlarging T2-hyperintense lesions (T2 lesions) as detected by brain Magnetic Resonance Imaging (MRI) during the double-blind period. * Number of new or enlarging T2 lesions by Week 96. * Annualized Relapse Rate (ARR) by Week 96. * Number of T1 gadolinium-enhancing (Gd+) lesions at 12 weeks. * Safety and tolerability profile (e.g., adverse events, infusion-related reactions).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard monitoring, with a specific focus on evaluating infusion-related reactions (IRRs) and possible infections. **Serious Adverse Events (SAEs):** Routine 24-hour reporting. Evaluation of severe IRRs, cardiac events, or serious infections. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee implemented to regularly review unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** \$N=171\$ to \$188\$ powered to detect noninferiority of ocrelizumab compared to fingolimod regarding ARR. **Handling of Missing Data:** Standard handling via Multiple Imputation and Mixed-Models for Repeated Measures (MMRM) for continuous secondary endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Onsite and remote monitoring visits to evaluate protocol adherence, capsule accountability, and proper weight-based IV dosing.

Audit Trail: Standardized eCRF locking and query resolution via data clarification forms.

11. Study Identifiers & Governance

Primary ID: WN42086 **Registry Number:** NCT05123703 **Posting ID:** 920486564724314556 **Sponsor/Lead Organization:** Roche **Study Title:** OPERETTA 2 **Official Regulatory Title:** A Phase III Multicenter, Randomized, Double-blind, Double-dummy Study to Evaluate Safety and Efficacy of Ocrelizumab in Comparison With Fingolimod in Children and Adolescents With Relapsing-Relmitting Multiple Sclerosis

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Relapsing-Relmitting Multiple Sclerosis (RRMS)

Diagnosis Threshold: Confirmed pediatric-onset RRMS (10 to < 18 years)

Medical Methodology: Disease-Modifying Therapy (DMT)

Optimization Target: Complete remission of clinical relapses and structural MRI activity **MeSH Code:** D020529 **SNOMED CT ID:** 426373005 **ATC Code:** L04AA27 (Fingolimod)

13. Study Procedures & Methodology

Management Pathway: Double-blind, double-dummy comparative pathway.

Education Protocol (HCP): Specialized training on weight-based pediatric dosing and monitoring for infusion-related reactions.

Education Protocol (Patient): Information given to children and their parents regarding daily oral capsule adherence and recognizing side effects.

Quality Audit Mechanism: Centralized blinded reading for MRI endpoints and formal confirmation of clinical relapses.

14. Observation & Follow-Up Schedule

Total Duration: Flexible double-blind period + at least 144 weeks

Open-Label Extension Onsite Visit Cadence: Baseline, Day 15, and every 24 weeks for IV infusions

Remote Monitoring Frequency: Every 4 weeks (Q4W) for ongoing safety and adherence checks

Checklist Review Interval: Every 12 weeks for thorough review of AEs and potential relapses

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					